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A Project Report on

Write Project Title Here

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A Report submitted to MIT Academy of Engineering, Alandi(D), Pune, An Autonomous Institute Affiliated to Savitribai Phule Pune University in partial fulfillment of the requirements of

SECOND YEAR BACHELOR OF TECHNOLOGY

in

Software Engineering

Department of Software Engineering

MIT Academy of Engineering

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Alandi (D), Pune – 412105

(2025–2026)

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CERTIFICATE

It is hereby certified that the work which is being presented in the Second Year Minor Project-Design Report entitled **"Write Project Title Here"**, in partial fulfillment of the requirements for the award of the Bachelor of Technology in Software Engineeringand submitted to the **Department of Software Engineering** of **MIT Academy of Engineering, Alandi(D), Pune, Affiliated to Savitribai Phule Pune University (SPPU), Pune**, is an authentic record of work carried out during Academic Year **2025–2026** Semester **V**, under the supervision of **Name of Project Guide**, **Department of Software Engineering**

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DECLARATION

We the undersigned solemnly declare that the project report is based on our own work carried out during the course of our study under the supervision of **Name of Project Guide**.

We assert the statements made and conclusions drawn are an outcome of our project work. We further certify that

1. The work contained in the report is original and has been done by us under the general supervision of our supervisor.
2. The work has not been submitted to any other Institution for any other degree/diploma/certificate in this Institute/University or any other Institute/University of India or abroad.
3. We have followed the guidelines provided by the Institute in writing the report.
4. Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them in the text of the report and giving their details in the references.

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Abstract

Write your abstract here

Acknowledgment

Write your acknowledgement here

students

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Chapter 1

Introduction

1.1 Background

Innovative improvements in the course of the most recent decade have yielded a few particular embedded platforms equipped for detecting, computing and communication. An immediate outcome of these improvements has been the development of, and consequent substantial scale research and engineering center in, the region of sensor systems. A sensor network is a system of embedded gadgets (sensor hubs) working in a co-agent way to detect and gather information for application-particular analysis. A sensor network application has a few plan measurements, detecting methodology (kind of sensor to utilize), calculation, communication and storage capacities, cost and size of every node, kind of energy source, design for organization, conventions for information scattering and communication, applications and administration devices, to give some examples.

A common and generally sent application classification is one that utilizations battery-controlled sensor networks. A couple of instantiations of such arrangements are for applications, for example, territory monitoring, environment observing in untamed life living spaces like Great Duck Island and James Reserve, to ponder climate conditions and creature transient examples, spring of gushing lava monitoring network of seismometers utilized as a part of a period synchronized and occasion activated way to gauge volcanic action, structural monitoring, a sensor system to identify in-

coming trains and measure the constrained and free vibrations of bridge; and vehicle following sensors put along or on streets, and on vehicles, recognize street, movement and ecological conditions-data which can be utilized to caution drivers of possibly hazardous conditions or on the other hand to illuminate about activity conditions for better course arranging (Ackermann & Utkin, 1998).

As found in applications said above, battery-controlled untethered sensor hubs can be utilized to instantiate applications in remote difficult to-achieve areas and areas where electrical cables don't exist. Further, being untethered, situation of these hubs is essentially unlimited and their development intemperate. A tradeoff of these advantages is the limited battery capacity nodes will work for a limited time frame, that is, just as long as the battery keeps going. Limited hub lifetime infers limited lifetime of the applications or, extra cost and unpredictability to routinely change batteries. The hubs could utilize expansive batteries for longer lifetimes, yet should manage expanded size, weight and cost. Hubs may likewise pick to utilize low-control equipment like low-control processor and radio, at the cost of lesser calculation capacity and lower transmission ranges, separately.

An elective procedure that has been connected to address the issue of limited hub lifetime is the utilization of energy gathering. Energy collecting alludes to bridling energy from the earth or other energy sources (body warm, foot strike, finger strokes) and changing over it to electrical energy. The outfit electrical energy controls the sensor hubs. Further, in light of the periodicity and size of harvestable energy, framework parameters of a hub can be tuned to expand hub and system execution. Therefore, energy collecting systems can possibly address the tradeoff between execution parameters what's more, lifetime of sensor hubs. The test lies in assessing the periodicity and greatness of the harvestable source and choosing which parameters to tune and all the while maintaining a strategic distance from untimely energy exhaustion before the following revive cycle.

Advances in energy scavenging materials has empowered the utilization of energy collecting gadgets.The gadgets can be worked utilizing low surrounding energy, for example, collected from indoor lights (J.-H. She, Kobayashi, Ohyama, & Xin, 2004).

A stage design joining energy from sun based, warm, and vibration sources is exhibited. The framework executed comprises of a reconfigurable multi-input, multi-yield switch network that joins energy from three unmistakable energy collecting sources photovoltaic, thermoelectric, and piezoelectric (J. She, Zhang, Lai, & Wu, 2012).(Young, 1993)

Energy harvesters are circuits which catches diminutive measure of energy from normally accessible sources, for example, sun oriented, warm, wind, vibrations, acoustic and magnetic field energy (Utkin, 1992). Numerous viable frameworks are accessible that reap energy from a solitary source. The test in such frameworks is to give a continuous power supply to the load influenced by dubious nature of these energy sources. To enhance the general unwavering quality and power of the framework, energy from numerous sources ought to be gathered.

The disturbances from changes in load and variation in source will rely upon numerous parameters of nature. To mitigate the impact of these disturbances, DC–DC converter system with robust control is incorporated as a scavenger circuit. A DC–DC converter is a nonlinear system, having precarious zero dynamics; sliding mode control (SMC) is determined to bargain the opposite impacts because of large variations in load.

DC–DC converters assume an imperative part in giving a consistent output voltage regardless of the input dynamics. A summed up strategy for modeling DC–DC converters was proposed in, that uses state space approach for two-state exchanging converters. The security, pick up and effectiveness of buck, boost and buck-boost converters is subject to switching time and duty cycle.

A new control method is used for energy scavenging and is shown in Figure 1.1. The concept is based on disturbance rejection leveraging the benefits of uncertainty and disturbance estimation. The proposed energy harvester scheme consists of an *Estimator* for estimating the uncertainty in inputs and disturbance in the form of load change, *Observer* for estimating the states (voltage and current), and a *Control law* based on *Sliding Mode Control*. This is motivated by the requirements of the practical plant (DC-DC converter); since most converter systems encountered in

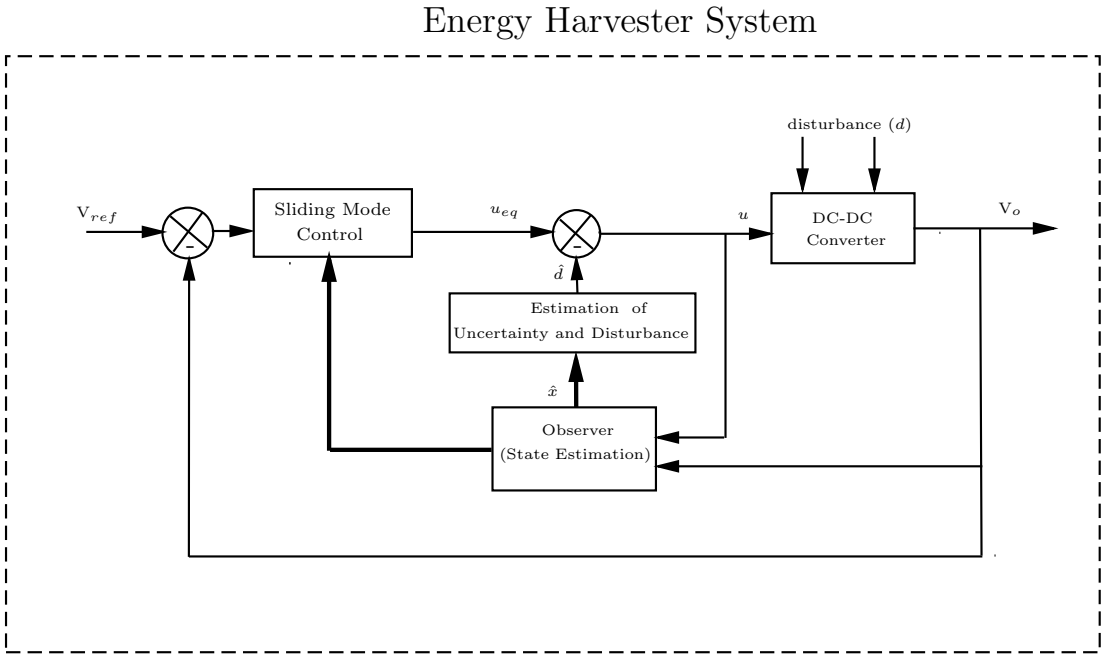


Figure 1.1: Configuration of Proposed Energy Harvester System

practice possess significant uncertainty (variable input) and are subjected to varied disturbances (change in load resistance). The robust control law is designed for scavenging uncertain energy and the sliding surface is designed to interdict for smooth control. The designed control laws are tested for various plants like buck, boost and buck-boost converter. The same is validated through simulation and hardware for handling uncertain inputs, variable loads and any external disturbances.

This work attempts to propose solution to address the limitations in the existing framework.

1.2 Motivation

1.3 Project Idea

1.4 Proposed Solution

1.5 Project Report Organization (Chapter wise summary)

Chapter 2

Literature Review

2.1 Related work And State of the Art (Latest work)

When the switch is opened i.e. in OFF state, the diode provides a path for the inductor current. The voltage across switch is $(V_i + V_o)$. The voltage across the inductor is $-V_o$ and causes the inductor current to decrease linearly with the slope of $-V_o/L$.

$$\Delta i_{L(off)} = \int_{t_{on}}^{t_{off}} \frac{V_L}{L} dt \tag{2.1}$$

$$\Delta i_{L(off)} = -\frac{V_o}{L} t_{off} \tag{2.2}$$

Since the duration of time when the switch OFF is given as, $t_{off} = (1 - D)\, T$. Then the equation (2.2) is written as,

$$\Delta i_{L(off)} = -\frac{V_o}{L}(1 - D)\, T \tag{2.3}$$

For steady state operation, the net change in inductor current must be zero over one period of time.

$$\Delta i_{L(on)} + \Delta i_{L(off)} = 0 \tag{2.4}$$

Substituting equation (**??**) and (2.3), into equation (2.4) and on simplifying we get,

$$\frac{V_o}{V_i} = \frac{D}{1 - D} \tag{2.5}$$

2.2 Limitation of State of the Art techniques

2.3 Discussion and future direction

2.4 Concluding Remarks

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Problem Definition and Scope

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Proposed Methodology

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5.4 Approach

Chapter 6

Conclusion

6.1 Conclusion

6.2 Future Scope

Appendices

Appendix A

Laplace Transform

The Laplace transform method is an operational method that can be used for solving linear differential equations. Laplace transforms can be used to convert many common functions, such as sinusoidal functions, damped sinusoidal functions, exponential functions, into algebraic functions of a complex variable s. Operations such as differentiation and integration can be replaced by algebraic operations in the complex plane. Thus, a linear differential equation can be transformed into an algebraic equation in a complex variable s. If the algebraic equation in s is solved for the dependent variable, then the solution of differential equation (*the inverse Laplace transform of the dependent variable*) may be found.

An advantage of the Laplace transform method is that, when the differential equation is solved, both the transient component and steady-state component of the solution can be obtained simultaneously.

Laplace Transform Pairs

f(t)	**F(s)**	**f(t)**	**F(s)**
$\delta(t)$	1	(step) 1	$\dfrac{1}{s}$
(ramp) t	$\dfrac{1}{s^2}$	(parabola) $\dfrac{t^2}{2}$	$\dfrac{1}{s^3}$
t^n	$\dfrac{n!}{s^{n+1}}$	e^{-at}	$\dfrac{1}{s+a}$
$t\,e^{-at}$	$\dfrac{1}{(s+a)^2}$	$t^n\,e^{-at}$	$\dfrac{n!}{(s+a)^{n+1}}$
$\sin\omega t$	$\dfrac{\omega}{s^2+\omega^2}$	$\cos\omega t$	$\dfrac{s}{s^2+\omega^2}$
$e^{-at}\,\sin\omega t$	$\dfrac{\omega}{(s+a)^2+\omega^2}$	$e^{-at}\,\cos\omega t$	$\dfrac{s}{(s+a)^2+\omega^2}$
$\dfrac{1}{b-a}\,\left(e^{-at}-e^{-bt}\right)$	$\dfrac{1}{(s+a)\,(s+b)}$	$\dfrac{1}{b-a}\,\left(b\,e^{-bt}-a\,e^{-at}\right)$	$\dfrac{s}{(s+a)\,(s+b)}$
$\dfrac{1}{a}\,\left(1-e^{-at}\right)$	$\dfrac{1}{s\,(s+a)}$		

Appendix B

Sponsorship Certificate

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